

Task 14: Interpret Each Evaluation Metric and Explain What It Reveals About Model Performance

1. Mean Squared Error (MSE)

Interpretation:

Mean Squared Error (MSE) measures the average of the squared differences between the actual and predicted house prices. A lower MSE indicates that the model's predictions are closer to the actual values. Since the errors are squared, larger errors receive a higher penalty.

2. Mean Absolute Error (MAE)

Interpretation:

Mean Absolute Error (MAE) calculates the average absolute difference between the actual and predicted values. It is easy to understand because it represents the average prediction error in the same unit as the target variable (INR). A smaller MAE indicates better model accuracy.

3. Root Mean Squared Error (RMSE)

Interpretation:

Root Mean Squared Error (RMSE) is the square root of MSE and measures the average prediction error in the same unit as the target variable. Lower RMSE values indicate better model performance. RMSE is more sensitive to large prediction errors than MAE.

4. R^2 Score (Coefficient of Determination)

Interpretation:

The R^2 Score indicates how well the independent variable explains the variation in the dependent variable. Its value ranges from **0 to 1**, where a value closer to **1** indicates a better-fitting model. For example, an R^2 score of **0.90** means that **90% of the variation in house prices is explained by the model**.

5. Adjusted R^2 Score

Interpretation:

Adjusted R^2 is an improved version of the R^2 Score that considers the number of independent variables used in the model. It penalizes unnecessary features that do not improve prediction accuracy. A higher Adjusted R^2 indicates a more reliable and efficient regression model.

Task 17: Explanation of Performance Change

◇ Why Performance Improves

- **More Predictors:** Multiple Linear Regression (MLR) includes several independent variables such as area, bedrooms, bathrooms, location, age, and lot size. This helps the model capture more real-world factors influencing house prices.
- **Higher R^2 / Adjusted R^2 :** With meaningful predictors, the model explains a larger portion of variance in house prices, resulting in a better fit.
- **Lower Errors (MSE, MAE, RMSE):** Predictions become closer to actual values because the model considers multiple dimensions of housing characteristics.

◇ Why Performance May Degrade

- **Irrelevant Predictors:** Adding features that do not truly affect house price introduces noise and reduces accuracy.
- **Multicollinearity:** Strong correlation between predictors (e.g., bedrooms and bathrooms) can distort coefficient estimates.
- **Overfitting:** Too many variables may fit the training data too closely, reducing generalization to new data.
- **Adjusted R^2 Drop:** If predictors are weak or redundant, Adjusted R^2 decreases, indicating unjustified model complexity.

Task 20: Identify Overfitting or Underfitting

◇ Overfitting

- Polynomial regression with higher degrees (e.g., degree 5 or more) fits the training data too closely, producing a **wiggly curve** that captures noise instead of the true pattern.
- R^2 is very high on the training dataset but drops significantly on the test dataset.
- This indicates poor generalization — the model performs well on known data but fails on unseen data.

◇ Underfitting

- A very simple model (like a straight line in Linear Regression) fails to capture the underlying relationship when the data is nonlinear.
- Both training and test errors remain high.
- This shows the model is too simplistic and cannot explain the variance in the data.

Task 21: Conceptual Explanation of Gradient Descent

- **Definition:** Gradient Descent is an optimization algorithm used to minimize a cost function (such as Mean Squared Error in regression).
- **Working Principle:** It iteratively adjusts parameters (weights) in the opposite direction of the gradient of the cost function.
- **Learning Rate (α):** Controls the step size.
 - Too high $\rightarrow$ overshooting, unstable convergence.
 - Too low $\rightarrow$ very slow convergence.
- **Goal:** Find parameter values that minimize the cost function, leading to the best model fit.

Task 25: Compare Convergence & Training Time

- **Batch Gradient Descent:**
 - Uses the entire dataset per update.
 - Converges smoothly but is computationally slow for large datasets.
- **Stochastic Gradient Descent (SGD):**

- Updates parameters for each sample individually.
- Fast updates, but the convergence path is noisy.
- Useful for very large datasets.
- **Mini-Batch Gradient Descent:**
 - Uses small batches (e.g., 32 samples).
 - Faster than Batch, smoother than SGD.
 - Provides the best balance of speed and stability.
 - Most commonly used in practice, especially in deep learning.